End-to-End E-commerce Intelligence System: Building a Customer 360 Analytics Framework

Project Report Submitted by: Rutesha

1. Business Problem Statement

In modern e-commerce ecosystems, data is generated across multiple independent systems such as customer management, order processing, payments, product catalogs, seller networks, and customer feedback platforms. These datasets are typically fragmented and stored in separate tables, making it difficult to extract unified insights.

The objective of this project is to simulate a real-world data analytics scenario with the following goals:

- Integrate multiple data sources into a unified analytical dataset
- Construct a Customer 360 View
- Analyze customer behavior, revenue patterns, and operational performance
- Identify key drivers of business growth and customer satisfaction
- Generate actionable, data-driven business recommendations

2. Dataset Description

The project uses a multi-table e-commerce dataset consisting of 8 relational CSV files. The orders table acts as the central fact table connecting all other datasets.

Table	Description	Records
customers.csv	Customer demographic and location information	99,441
orders.csv	Central order lifecycle table with timestamps	99,441
order_item.csv	Product-level details for each order	112,650
payments.csv	Payment type and value per order	103,886
reviews.csv	Customer feedback and review scores	99,224
products.csv	Product details and categories	32,951
sellers.csv	Seller location information	3,095
category_translation.csv	Maps Portuguese categories to English	71

The tables are joined using the following keys:

- order_id connects orders, order_item, payments, and reviews
- customer_id connects orders and customers
- product_id connects order_item and products
- seller_id connects order_item and sellers

- product_category_name connects products and category_translation

3. Methodology

Step 1: Data Loading and Exploration

All 8 datasets were loaded using the Pandas library. Each table was inspected using head(), info(), describe(), and isnull() functions to understand structure, data types, and missing values.

Step 2: Data Cleaning and Preprocessing

The following cleaning operations were performed on each table:

- Missing values were handled based on count. Values below 1000 were filled using appropriate strategies such as median for numeric columns and placeholder text for string columns. Columns with more than 1000 missing values were dropped.
- Duplicate records were removed from all tables.
- Date columns were converted from object type to datetime format across orders, order_item, and reviews tables.
- Data ranges were validated for price, freight value, payment value, and review scores.
- All column names were standardized to lowercase with underscores.

Table	Column	Missing Count	Action Taken
orders	order_approved_at	160	Filled with order_purchase_timestamp
orders	order_delivered_carrier_date	1,783	Rows dropped
orders	order_delivered_customer_date	2,965	Rows dropped
reviews	review_comment_title	87,656	Rows dropped
reviews	review_comment_message	58,247	Rows dropped
products	product_category_name	610	Filled with unknown
products	Numeric columns	2 to 610	Filled with median

Step 3: Data Integration

A Master Dataset was constructed by merging all 8 tables using the recommended sequence. GroupBy and aggregation was applied before merging tables with one-to-many relationships to avoid row duplication.

Merge	Tables	Key	Approach
1	orders + customers	customer_id	Direct merge
2	orders + order_item	order_id	GroupBy and Agg (one-to-many)
3	order_item + products	product_id	Direct merge
4	orders + payments	order_id	GroupBy and Agg (one-to-many)
5	orders + reviews	order_id	GroupBy and Agg (one-to-many)
6	order_item + sellers	seller_id	Direct merge
7	products + category_translation	product_category_name	Direct merge

The final Master Dataset contains 42 columns combining all tables into a single unified business view.

Step 4: Feature Engineering

Feature	Formula / Method
total_order_value	total_price + total_freight per order
delivery_time_days	order_delivered_customer_date minus order_purchase_timestamp
total_items	Count of items per order
purchase_frequency	Number of unique orders per customer
customer_lifetime_value	Sum of total_order_value per customer
avg_order_value	Mean of total_order_value per customer
customer_segment	Low, Mid, High Value based on CLV percentiles (0.33 and 0.66)
order_purchase_year/month/hour/day	Extracted from order_purchase_timestamp

Step 5: Exploratory Data Analysis

Structured analysis was performed across five dimensions: Customer Analysis, Revenue and Order Analysis, Product Analysis, Seller Analysis, and Review and Satisfaction Analysis.

Step 6: Data Visualization

Visualizations were created using Matplotlib and Seaborn including time series plots for sales trends, bar charts for category and state performance, histograms for distribution analysis, box plots for outlier detection, and heatmaps for correlation and seasonal patterns.

4. Key Findings

Customer Analysis

Metric	Value
Total Unique Customers	93,355
New Customers (one-time buyers)	90,554 (97.0%)
Repeat Customers	2,801 (3.0%)
Average Purchase Frequency	1.08 orders per customer
Maximum Purchase Frequency	15 orders by one customer
Peak Shopping Hour	16:00 (4 PM)
Peak Shopping Day	Monday

Customer Segment Breakdown:

Segment	Customers	Revenue	Revenue %	Avg Order Value
High Value	23,490	R$ 13,252,300	67.03%	R$ 353.71
Mid Value	33,773	R$ 4,544,981	22.99%	R$ 124.99
Low Value	36,092	R$ 1,974,656	9.99%	R$ 54.30

Revenue and Order Analysis

Metric	Value
Total Revenue Generated	R$ 19,771,937.90
Total Orders	96,475
Average Order Value	R$ 179.43
Peak Revenue Month	May

Revenue by Payment Type:

Payment Type	Total Revenue	Total Orders	Revenue %
Credit Card	R$ 15,207,234	73,218	76.9%
Boleto	R$ 3,943,080	19,191	19.9%
Voucher	R$ 374,806	2,582	1.9%
Debit Card	R$ 246,385	1,483	1.2%

Top 5 States by Revenue:

State	Revenue	Revenue %
SP	R$ 7,402,533	37.44%
RJ	R$ 2,689,035	13.60%
MG	R$ 2,280,614	11.53%
RS	R$ 1,110,441	5.62%
PR	R$ 1,030,030	5.21%

Product Analysis

Metric	Value
Total Product Categories	71
Total Unique Products	32,213
Top Revenue Category	bed_bath_table (8.56%)

Top 5 Revenue Driving Categories:

Category	Revenue	Revenue %
bed_bath_table	R$ 1,691,702	8.56%
health_beauty	R$ 1,621,265	8.20%
computers_accessories	R$ 1,549,136	7.84%
furniture_decor	R$ 1,393,752	7.05%
watches_gifts	R$ 1,386,828	7.01%

Seller Analysis

Metric	Value
Total Sellers	2,970
Top Seller State	SP (1,849 sellers)
Top Seller Revenue	R$ 505,410

Review and Satisfaction Analysis

Metric	Value
Average Review Score	3.79 out of 5
Median Review Score	5.00
Dissatisfaction Rate (score 1 or 2)	2.0%
Total Orders Delayed	34,590 (35.9%)
Average Delivery Time	12.0 days

Metric	Value
Maximum Delivery Time	209 days

Average Delivery Time by Review Score:

Review Score	Average Delivery Time
Score 1 (Worst)	13.89 days
Score 2	11.42 days
Score 3	10.90 days
Score 4	10.32 days
Score 5 (Best)	8.53 days

Top 5 Categories with Low Ratings (score 2 or below):

Category	Low Rating Count
bed_bath_table	221
watches_gifts	201
health_beauty	179
computers_accessories	144
housewares	138

5. Business Insights

Insight 1: Customer Retention Crisis 97 percent of all customers make only one purchase and never return. The average purchase frequency is just 1.08 orders per customer which is critically low. The industry average repeat purchase rate is between 20 and 30 percent, indicating a serious gap in retention strategy.

Insight 2: High Value Customers Drive Revenue High Value customers make up only 25 percent of the total customer base but contribute 67 percent of total revenue. Their average order value of R$ 353.71 is more than 6 times higher than Low Value customers at R$ 54.30. Retaining and growing this segment is the highest priority for revenue growth.

Insight 3: Delivery Time Directly Impacts Ratings There is a clear and direct relationship between delivery time and review scores. Customers who received their orders in 8.5 days gave 5 stars while customers who received in 13.9 days gave 1 star. A 5.36 day difference separates the best and worst rated experiences. Reducing delivery time is the most effective lever for improving customer satisfaction.

Insight 4: High Order Delay Rate 35.9 percent of all orders were delivered late meaning 1 in every 3 customers received a delayed order. The industry benchmark is below 10 percent. The maximum recorded delivery time of 209 days is an extreme outlier requiring immediate investigation. Combined with the 97 percent one-time buyer rate, each delayed order represents a permanently lost customer.

Insight 5: Revenue Concentration Risk The top 5 product categories contribute 38.66 percent of total revenue and the top 3 states contribute 62.57 percent of revenue. This concentration creates business risk if performance drops in any key category or state.

Insight 6: Top Revenue Categories Have Quality Issues All 5 of the top dissatisfied categories are also in the top 5 revenue categories. bed_bath_table generates the highest revenue at 8.56 percent but also has the most low ratings at 221. This combination of high revenue and high dissatisfaction is a warning sign that quality issues could erode top revenue categories over time.

Insight 7: Credit Card Dominates Payments Credit card accounts for 76.9 percent of total revenue. Debit card and voucher together contribute only 3.1 percent indicating underdeveloped alternative payment channels that could attract new customer segments.

6. Recommendations

Recommendation 1: Solve the Customer Retention Crisis

Priority: Critical. Action required within 30 days.

Problem: 97 percent of customers buy only once. The business is failing at retention.

- Send automated follow-up email 7 days after delivery with a 10 percent discount for next purchase
- Launch a loyalty points program where every R$ 10 spent earns 1 redeemable point
- Create a referral program giving existing customers R$ 20 voucher for every new customer they refer
- For customers inactive for 60 or more days send a win-back campaign with 15 percent discount
- Introduce subscription model for frequently bought categories like health_beauty and bed_bath_table

Expected Result: Increase repeat purchase rate from 3 percent to 10 to 15 percent, adding R$ 2 to 3 million in revenue.

Recommendation 2: Reduce Delivery Time from 12 to 8 Days

Priority: Critical. Action required within 3 months.

Problem: Average delivery of 12 days is too high. Every extra day of delivery results in a lower review score.

- Set a mandatory delivery SLA of 8 days for all orders
- Build regional fulfillment centers in SP, RJ and MG to reduce last mile delivery by 3 to 4 days
- Partner with express logistics providers for orders above R$ 200
- Add real-time delivery tracking for customers
- Investigate the 209 day delivery outlier immediately and eliminate the root cause

Expected Result: Reduce average delivery from 12 to 8 days and improve average review score to 4.5 or above.

Recommendation 3: Reduce 35.9 Percent Order Delay Rate

Priority: Critical. Action required within 3 months.

Problem: 34,590 out of 96,475 orders are delayed. Industry benchmark is below 10 percent.

- Build an automated system that flags orders at risk of delay 24 hours before estimated delivery date
- Send proactive delay notifications to customers as soon as delay is detected
- Introduce delay compensation: 5 percent voucher for 1 to 3 day delay, 10 percent for 3 to 7 days, full refund option for 7 or more days
- Set financial penalties for logistics partners who cause delays

Expected Result: Reduce delay rate from 35.9 percent to below 15 percent.

Recommendation 4: Launch VIP Program for High Value Customers

Priority: High. Action required within 3 months.

Problem: High Value customers contribute 67 percent of revenue but receive no special treatment.

- Launch exclusive VIP membership with free express delivery and priority customer support
- Give early access to new products and flash sales
- Send personalized birthday and anniversary offers with exclusive discounts
- Assign dedicated account managers for the top 500 customers by lifetime value

Expected Result: Increase High Value segment revenue by 20 to 25 percent.

Recommendation 5: Fix Quality in Top Dissatisfied Categories

Priority: High. Action required within 3 months.

Problem: bed_bath_table, watches_gifts and health_beauty are both top revenue and top dissatisfied categories, putting R$ 7.6 million of revenue at risk.

- Conduct immediate seller quality audit for bed_bath_table and watches_gifts
- Set mandatory product quality standards before any new listing in top 5 categories
- Remove sellers with average rating below 2.5 from these categories immediately
- Add verified buyer photos and detailed product descriptions to reduce expectation mismatch
- Introduce buyer protection with full refund if rating is below 2 and free returns within 30 days

Expected Result: Reduce low ratings by 40 to 50 percent in top revenue categories.

Recommendation 6: Focus on Top 5 Revenue Categories

Priority: High. Action required within 6 months.

Problem: Only 5 categories drive nearly 40 percent of revenue while other categories underperform.

- Increase product variety in top 5 categories by recruiting more sellers
- Run targeted promotions and bundle offers for these categories
- Feature top categories prominently on the homepage and in marketing campaigns

Expected Result: 10 to 15 percent revenue increase from top 5 categories.

Recommendation 7: Reduce Geographic Revenue Concentration

Priority: Medium. Action required within 6 months.

Problem: SP alone contributes 37.44 percent of revenue creating concentration risk.

- Set up fulfillment centers in SP, RJ and MG to serve high demand regions faster
- Launch targeted marketing campaigns in RS and PR
- Run state-specific discount campaigns in bottom 10 revenue states for 3 months
- Recruit local sellers in low revenue states to improve product availability

Expected Result: Reduce SP dependency from 37 percent to 30 percent and grow low-performing states by 20 percent.

Recommendation 8: Leverage Peak Shopping Times

Priority: Medium. Action required within 6 months.

Problem: Peak shopping pattern of Monday at 4 PM is not being used for targeted campaigns.

- Schedule all promotional emails and push notifications to go out at 3:30 PM every Monday

- Run Monday-only flash sales from 4 PM to 7 PM
- Ensure maximum inventory availability on Monday afternoons
- Schedule social media ads between 3 PM and 6 PM on Mondays

Expected Result: 10 to 15 percent increase in Monday afternoon sales.

7. Priority Action Plan

Priority	**Timeframe**	**Action**
Critical	Within 30 days	Launch loyalty program and post-purchase emails for retention
Critical	Within 30 days	Investigate and fix 209 day delivery outlier
Critical	Within 30 days	Audit sellers in bed_bath_table category
High	Within 3 months	Reduce average delivery from 12 to 8 days
High	Within 3 months	Reduce delay rate from 35.9 to below 15 percent
High	Within 3 months	Launch VIP program for 23,490 High Value customers
Medium	Within 6 months	Geographic expansion in low revenue states
Medium	Within 6 months	Monday 4 PM peak time campaign strategy
Low	Within 12 months	Add new payment options and expand seller network

8. Estimated Financial Impact

Recommendation	**Estimated Revenue Gain**
Recommendation 1 - Customer Retention Fix	+ R$ 2.5 million
Recommendation 2 - Delivery Improvement	+ R$ 1.5 million
Recommendation 3 - Delay Reduction	+ R$ 1.0 million
Recommendation 4 - VIP Program	+ R$ 3.3 million
Recommendation 6 - Top Category Focus	+ R$ 2.0 million
Total Potential Gain	+ R$ 10.3 million

Conclusion: The business currently generates R$ 19.77 million in revenue with 96,475 orders and 93,355 customers. The two most critical problems are a 97 percent one-time buyer rate and a 35.9 percent order delay rate. Addressing these two issues alone has the potential to add R$ 5 million or more in additional revenue and significantly improve customer satisfaction scores across all dimensions.